# Sec 1

**Directions for questions 1 to 4:** Answer the questions on the basis of the information given below.

Women's Cricket Teams from 7 countries – A, B, C, D, E, F, and G participated in the recent WT-20 Asia Cup. The tournament was played at the XYZ International Cricket Stadium.
All the 7 teams played in a round-robin stage, each team initially played once with each team, with the top 4 progressing to the semi-finals. If the number of points is equal, then the Net Run Rate (NRR) will be considered and the team with a higher NRR will qualify for semi-finals with higher seeding.
Out of the top 4 teams, the first semi-final match was played between the top scorer (as per total points awarded) and the lowest scorer (as per total points awarded); and the second semi-final match was played between the other two teams. The winners of the two semi-finals played in the final match and the winner got the Asia Cup Trophy. No match was drawn/tie in either the semi-finals or the final. Two points were awarded for a win, 1 for a draw, and 0 for a loss for every match.
The table given below shows partial information about matches played in the Round-robin stage.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | | 1 | 0 | | > 3 |
| B | | | | 10 | > 1 but < 2 |
| C | | 2 | 0 | | |
| D | | | 0 | 6 | |
| E | | | | 5 | < 0.5 |
| F | 1 | | | 3 | |
| G | 0 | | 0 | | < –3 |

**Q.1 [11831809]**
What is the maximum total number of drawn matches that could have been played in the tournament?

**Solution:**
**Correct Answer : 2**

Answer key/Solution

Team A won 5 matches. So the points awarded to team A were 10. Team C won 4 matches. So the points awarded to team C were 8. Total points awarded to team D were 6. So team D won 3 matches and lost 3 matches.
Team G lost all matches. So total points awarded to team G was 0. Team F awarded 3 points. So team F lost 4 matches and 1 match was drawn.
The total points awarded to Team B were 10. So team B won 5 or 4 matches. And either 0 or 2 matches were drawn.
Total points awarded to team E were 5. So team E won 2 matches and 1 match was drawn.
The given information can be shown in the following table.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | 5 | 1 | 0 | 10 | > 3 |
| B | 4/5 | 0/1 | 2/0 | 10 | > 1 but < 2 |
| C | 4 | 2 | 0 | 8 | |
| D | 3 | 3 | 0 | 6 | |
| E | 2 | 3 | 1 | 5 | < 0.5 |
| F | 1 | 4 | 1 | 3 | |
| G | 0 | 6 | 0 | 0 | < –3 |

**The maximum total number of drawn matches that can be played in the tournament is (1 + 1 + 2)/2 = 2.**

Bookmark FeedBack

**Directions for questions 1 to 4:** Answer the questions on the basis of the information given below.

Women's Cricket Teams from 7 countries – A, B, C, D, E, F, and G participated in the recent WT-20 Asia Cup. The tournament was played at the XYZ International Cricket Stadium.
All the 7 teams played in a round-robin stage, each team initially played once with each team, with the top 4 progressing to the semi-finals. If the number of points is equal, then the Net Run Rate (NRR) will be considered and the team with a higher NRR will qualify for semi-finals with higher seeding.
Out of the top 4 teams, the first semi-final match was played between the top scorer (as per total points awarded) and the lowest scorer (as per total points awarded); and the second semi-final match was played between the other two teams. The winners of the two semi-finals played in the final match and the winner got the Asia Cup Trophy. No match was drawn/tie in either the semi-finals or the final. Two points were awarded for a win, 1 for a draw, and 0 for a loss for every match.
The table given below shows partial information about matches played in the Round-robin stage.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | | 1 | 0 | | > 3 |
| B | | | | 10 | > 1 but < 2 |
| C | | 2 | 0 | | |
| D | | | 0 | 6 | |
| E | | | | 5 | < 0.5 |
| F | 1 | | | 3 | |
| G | 0 | | 0 | | < –3 |

**Q.2 [11831809]**
The second semi-final match was played between which of the following two teams?

1 ◯ A & C

2 ◯ B & D

3 ◯ B & C

4 ◯ A & B

**Solution:**
**Correct Answer : 3**

Answer key/Solution

Team A won 5 matches. So the points awarded to team A were 10. Team C won 4 matches. So the points awarded to team C were 8. Total points awarded to team D were 6. So team D won 3 matches and lost 3 matches.
Team G lost all matches. So total points awarded to team G was 0. Team F awarded 3 points. So team F lost 4 matches and 1 match was drawn.
The total points awarded to Team B were 10. So team B won 5 or 4 matches. And either 0 or 2 matches were drawn.
Total points awarded to team E were 5. So team E won 2 matches and 1 match was drawn.
The given information can be shown in the following table.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | 5 | 1 | 0 | 10 | > 3 |
| B | 4/5 | 0/1 | 2/0 | 10 | > 1 but < 2 |
| C | 4 | 2 | 0 | 8 | |
| D | 3 | 3 | 0 | 6 | |
| E | 2 | 3 | 1 | 5 | < 0.5 |
| F | 1 | 4 | 1 | 3 | |
| G | 0 | 6 | 0 | 0 | < –3 |

**Total points awarded to teams A and B is equal i.e., 10 points to each team. Net Run Rate (NRR) of Team A is more than that of Team B. Hence, teams A & D were played in first semi-final match and teams B & C were played in the second semi-final match.**

Bookmark FeedBack

**Directions for questions 1 to 4:** Answer the questions on the basis of the information given below.

Women's Cricket Teams from 7 countries – A, B, C, D, E, F, and G participated in the recent WT-20 Asia Cup. The tournament was played at the XYZ International Cricket Stadium.
All the 7 teams played in a round-robin stage, each team initially played once with each team, with the top 4 progressing to the semi-finals. If the number of points is equal, then the Net Run Rate (NRR) will be considered and the team with a higher NRR will qualify for semi-finals with higher seeding.
Out of the top 4 teams, the first semi-final match was played between the top scorer (as per total points awarded) and the lowest scorer (as per total points awarded); and the second semi-final match was played between the other two teams. The winners of the two semi-finals played in the final match and the winner got the Asia Cup Trophy. No match was drawn/tie in either the semi-finals or the final. Two points were awarded for a win, 1 for a draw, and 0 for a loss for every match.
The table given below shows partial information about matches played in the Round-robin stage.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | | 1 | 0 | | > 3 |
| B | | | | 10 | > 1 but < 2 |
| C | | 2 | 0 | | |
| D | | | 0 | 6 | |
| E | | | | 5 | < 0.5 |
| F | 1 | | | 3 | |
| G | 0 | | 0 | | < –3 |

**Q.3 [11831809]**
Which of the following statements must be false?

1 ◯ The match between Teams E and F was not drawn in round-robin stage.

2 ◯ Team B played more than two drawn matches in round-robin stage.

3 ◯ Team D won the round-robin stage match against Team E.

4 ◯ Team A lost the round-robin stage match against Team B.

**Solution:**
**Correct Answer : 2**

Answer key/Solution

Team A won 5 matches. So the points awarded to team A were 10. Team C won 4 matches. So the points awarded to team C were 8. Total points awarded to team D were 6. So team D won 3 matches and lost 3 matches.
Team G lost all matches. So total points awarded to team G was 0. Team F awarded 3 points. So team F lost 4 matches and 1 match was drawn.
The total points awarded to Team B were 10. So team B won 5 or 4 matches. And either 0 or 2 matches were drawn.
Total points awarded to team E were 5. So team E won 2 matches and 1 match was drawn.
The given information can be shown in the following table.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | 5 | 1 | 0 | 10 | > 3 |
| B | 4/5 | 0/1 | 2/0 | 10 | > 1 but < 2 |
| C | 4 | 2 | 0 | 8 | |
| D | 3 | 3 | 0 | 6 | |
| E | 2 | 3 | 1 | 5 | < 0.5 |
| F | 1 | 4 | 1 | 3 | |
| G | 0 | 6 | 0 | 0 | < –3 |

Here, W = won, L = lost and D = drawn
**Case 1:** The match between teams E and F was not drawn.

| Country | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| A | | L | W | W | W | W | W |
| B | W | | W | W | D | D | W |
| C | L | L | | W | W | W | W |
| D | L | L | L | | W | W | W |
| E | L | D | L | L | | W | W |
| F | L | D | L | L | L | | W |
| G | L | L | L | L | L | L | |

**Case 2:** The match between teams E and F was drawn.

| Country | A | B | C | D | E | F | G |
|---|---|---|---|---|---|---|---|
| A | | | | | | | W |
| B | | | | | | | W |
| C | | | | | | | W |
| D | | | | | | | W |
| E | | | | | | D | W |
| F | | | | | D | | W |
| G | L | L | L | L | L | L | |

The statement given in option (2) must be false.

Bookmark FeedBack

**Directions for questions 1 to 4:** Answer the questions on the basis of the information given below.

Women's Cricket Teams from 7 countries – A, B, C, D, E, F, and G participated in the recent WT-20 Asia Cup. The tournament was played at the XYZ International Cricket Stadium.
All the 7 teams played in a round-robin stage, each team initially played once with each team, with the top 4 progressing to the semi-finals. If the number of points is equal, then the Net Run Rate (NRR) will be considered and the team with a higher NRR will qualify for semi-finals with higher seeding.
Out of the top 4 teams, the first semi-final match was played between the top scorer (as per total points awarded) and the lowest scorer (as per total points awarded); and the second semi-final match was played between the other two teams. The winners of the two semi-finals played in the final match and the winner got the Asia Cup Trophy. No match was drawn/tie in either the semi-finals or the final. Two points were awarded for a win, 1 for a draw, and 0 for a loss for every match.
The table given below shows partial information about matches played in the Round-robin stage.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | | 1 | 0 | | > 3 |
| B | | | | 10 | > 1 but < 2 |
| C | | 2 | 0 | | |
| D | | | 0 | 6 | |
| E | | | | 5 | < 0.5 |
| F | 1 | | | 3 | |
| G | 0 | | 0 | | < –3 |

**Q.4 [11831809]**
What can be the maximum difference between the total points awarded to the teams that reaches in the final, at the end of the tournament?

1 ◯ 10

2 ◯ 4

3 ◯ 8

4 ◯ 6

**Solution:**
**Correct Answer : 4**

Answer key/Solution

Team A won 5 matches. So the points awarded to team A were 10. Team C won 4 matches. So the points awarded to team C were 8. Total points awarded to team D were 6. So team D won 3 matches and lost 3 matches.
Team G lost all matches. So total points awarded to team G was 0. Team F awarded 3 points. So team F lost 4 matches and 1 match was drawn.
The total points awarded to Team B were 10. So team B won 5 or 4 matches. And either 0 or 2 matches were drawn.
Total points awarded to team E were 5. So team E won 2 matches and 1 match was drawn.
The given information can be shown in the following table.

| Country | No. of matches won | No. of matches lost | No. of matches drawn | No. of points awarded | Net Run Rate |
|---|---|---|---|---|---|
| A | 5 | 1 | 0 | 10 | > 3 |
| B | 4/5 | 0/1 | 2/0 | 10 | > 1 but < 2 |
| C | 4 | 2 | 0 | 8 | |
| D | 3 | 3 | 0 | 6 | |
| E | 2 | 3 | 1 | 5 | < 0.5 |
| F | 1 | 4 | 1 | 3 | |
| G | 0 | 6 | 0 | 0 | < –3 |

**Teams A & D played in first semi-final match and teams B & C played in the second semi-final match. For the maximum difference, team D won the first semi-final and team B won the second semi-final. If team B won the trophy, then the total points awarded to team B = 10 + 2 + 2 = 14 and the total points awarded to team D = 6 + 2 = 8. Hence, the required maximum difference at the end of the tournament = 14 – 8 = 6.**

Bookmark FeedBack

**Directions for questions 5 to 10:** Answer the questions on the basis of the information given below.

Five aspirants - Aditi, Basil, Charu, Dharmu and Ekta appeared in five entrance tests - CET, SAT, GAT, IFT and MET - for admission in management colleges. Their marks in the above tests and their work experience while appearing for that test are as shown in the table given below:

| Name | Marks in test | | | | | Work experience (in years) while appearing | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CET | SAT | GAT | IFT | MET | CET | SAT | GAT | IFT | MET |
| Aditi | 38 | 64 | 51 | 76 | 82 | 1 | 2 | 2 | 1 | 1 |
| Basil | 42 | 56 | 57 | 60 | 86 | 1 | 2 | 3 | 1 | 4 |
| Charu | 40 | 62 | 63 | 62 | 88 | 2 | 1 | 2 | 2 | 3 |
| Dharmu | 45 | 60 | 60 | 68 | 85 | 1 | 1 | 1 | 4 | 2 |
| Ekta | 41 | 59 | 65 | 72 | 94 | 3 | 1 | 2 | 1 | 0 |

If an aspirant's marks in any test is equal to the average of the marks of two or three or four of the other aspirants and in addition his/her work experience while appearing for that test is the average of that of two or three or four of the other aspirants, then that aspirant will definitely receive a call letter for admission through that test. For the rest of the aspirants, the call letter may also based on other criteria apart from just their marks and the work experience.

**For example:-** If Aditi's marks in a test was an average of that of (2 others) Basil and Charu or (3 others) Basil, Charu and Dharmu or (4 others) Basil, Charu, Dharmu and Ekta and similarly work experience is also such an average, then Aditi must get a call letter to get admission through that test. If Aditi's marks does not satisfy this condition the other criteria will be considered for the call letter.

Some other known facts are as below:
(i) For every test there were either 3 or 4 aspirants who received a call letter.
(ii) Each aspirant received call letters through at least two tests.
(iii) Among all the five tests, only one aspirant with the lowest marks in a test received a call letter for admission.
(iv) No aspirant received call letters for admission through all five tests.
(v) If Aditi receives a call letter through a test, then Charu and Dharmu also receive call letters for the same test.
(vi) A total of 16 call letters were issued based on the above information.

**Q.5 [11831809]**
Through which of the following entrance tests did four aspirants receive call letters?

1 ◯ SAT

2 ◯ GAT

3 ◯ Either SAT or IFT

4 ◯ Either GAT or IFT

**Solution:**
**Correct Answer : 3**

Answer key/Solution

**Step 1:**
First of all, we would pick those who must have received admission call letter based on information given in the question (Average based condition for both marks and experience).
From this; we see that 8 call letters were issued as below:

| | CET | SAT | GAT | IFT | MET |
|---|---|---|---|---|---|
| Aditi | | | | | |
| Basil | 42 ✓ | | | | |
| Charu | 40 ✓ | 62 ✓ | | | 88 ✓ |
| Dharmu | | 60 ✓ | | | 85 ✓ |
| Ekta | | 59 ✓ | | 72 ✓ | |

**Step 2:**
From condition (vi), total 16 call letters were issued which is only possible in such a way that 4 letters through, 1 test and 3 through each remaining.
Now, from condition (iii),

**Case 1:**
Least scorer in CET i.e., Aditi must get call letter.
This case would not be possible because there would be 2 tests through which 4 call letters were issued which violates conditions (i) and (vi) together.
**Case 2:**
Least scorer in SAT i.e., Basil must get call letter.
This case is not possible as it violates condition (ii).
**Case 3:**
Least scorer in GAT i.e., Aditi must get call letter.
This case violates condition (v).
**Case 4:**
Least scorer in IFT i.e., Basil must get call letter.
This case also violates condition (v).
**Case 5:**
Least scorer in MET i.e., Aditi must get call letter. This case is possible.
**Step 3:**
From condition (iii), scorer of 38 in CET, 56 in SAT, 51 in GAT, 60 in IFT must not get call letter.
From conditions (i) and (v); we can conclude that Charu and Dharmu must get call letter through IFT. And; now Charu can't get call letter through GAT in order to satisfy condition (iv). We can also see that;
Basil, Dharmu and Ekta must get call letter GAT. Now; Dharmu can't get call letter through CET and Ekta must receive call letter through CET but not through MET.
If Basil gets call letter through MET, then Aditi can't get call letter through both SAT and IFT. But in this case condition (ii) is violated.
Therefore, Basil had not received call letter through MET. And; Aditi had received admission call letter through either SAT or IFT.
Hence, the information can be shown in the following table.

| Test / Name | CET | SAT | GAT | IFT | MET | Total |
|---|---|---|---|---|---|---|
| Aditi | 38 x | 64 ✓ / x | 51 x | 76 x / ✓ | 82 ✓ | 2 |
| Basil | 42 ✓ | 56 x | 57 ✓ | 60 x | 86 x | 2 |
| Charu | 40 ✓ | 62 ✓ | 63 x | 62 ✓ | 88 ✓ | 4 |
| Dharmu | 45 x | 60 ✓ | 60 ✓ | 68 ✓ | 85 ✓ | 4 |
| Ekta | 41 ✓ | 59 ✓ | 65 ✓ | 72 ✓ | 94 x | 4 |
| Total | 3 | 4/3 | 3 | 3/4 | 3 | 16 |

**Four aspirants had received call letter through either SAT or IFT.**

Bookmark FeedBack

**Directions for questions 5 to 10:** Answer the questions on the basis of the information given below.

Five aspirants - Aditi, Basil, Charu, Dharmu and Ekta appeared in five entrance tests - CET, SAT, GAT, IFT and MET - for admission in management colleges. Their marks in the above tests and their work experience while appearing for that test are as shown in the table given below:

| Name | Marks in test | | | | | Work experience (in years) while appearing | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CET | SAT | GAT | IFT | MET | CET | SAT | GAT | IFT | MET |
| Aditi | 38 | 64 | 51 | 76 | 82 | 1 | 2 | 2 | 1 | 1 |
| Basil | 42 | 56 | 57 | 60 | 86 | 1 | 2 | 3 | 1 | 4 |
| Charu | 40 | 62 | 63 | 62 | 88 | 2 | 1 | 2 | 2 | 3 |
| Dharmu | 45 | 60 | 60 | 68 | 85 | 1 | 1 | 1 | 4 | 2 |
| Ekta | 41 | 59 | 65 | 72 | 94 | 3 | 1 | 2 | 1 | 0 |

If an aspirant's marks in any test is equal to the average of the marks of two or three or four of the other aspirants and in addition his/her work experience while appearing for that test is the average of that of two or three or four of the other aspirants, then that aspirant will definitely receive a call letter for admission through that test. For the rest of the aspirants, the call letter may also based on other criteria apart from just their marks and the work experience.

**For example:-** If Aditi's marks in a test was an average of that of (2 others) Basil and Charu or (3 others) Basil, Charu and Dharmu or (4 others) Basil, Charu, Dharmu and Ekta and similarly work experience is also such an average, then Aditi must get a call letter to get admission through that test. If Aditi's marks does not satisfy this condition the other criteria will be considered for the call letter.

Some other known facts are as below:
(i) For every test there were either 3 or 4 aspirants who received a call letter.
(ii) Each aspirant received call letters through at least two tests.
(iii) Among all the five tests, only one aspirant with the lowest marks in a test received a call letter for admission.
(iv) No aspirant received call letters for admission through all five tests.
(v) If Aditi receives a call letter through a test, then Charu and Dharmu also receive call letters for the same test.
(vi) A total of 16 call letters were issued based on the above information.

**Q.6 [11831809]**
Which of the following is the marks of an aspirant who definitely receive a call letter?

1 ◯ 64

2 ◯ 76

3 ◯ 41

4 ◯ 60

**Solution:**
**Correct Answer : 3**

Answer key/Solution

**Step 1:**
First of all, we would pick those who must have received admission call letter based on information given in the question (Average based condition for both marks and experience).
From this; we see that 8 call letters were issued as below:

| | CET | SAT | GAT | IFT | MET |
|---|---|---|---|---|---|
| Aditi | | | | | |
| Basil | 42 ✓ | | | | |
| Charu | 40 ✓ | 62 ✓ | | | 88 ✓ |
| Dharmu | | 60 ✓ | | | 85 ✓ |
| Ekta | | 59 ✓ | | 72 ✓ | |

**Step 2:**
From condition (vi), total 16 call letters were issued which is only possible in such a way that 4 letters through, 1 test and 3 through each remaining.
Now, from condition (iii),

**Case 1:**
Least scorer in CET i.e., Aditi must get call letter.
This case would not be possible because there would be 2 tests through which 4 call letters were issued which violates conditions (i) and (vi) together.
**Case 2:**
Least scorer in SAT i.e., Basil must get call letter.
This case is not possible as it violates condition (ii).
**Case 3:**
Least scorer in GAT i.e., Aditi must get call letter.
This case violates condition (v).
**Case 4:**
Least scorer in IFT i.e., Basil must get call letter.
This case also violates condition (v).
**Case 5:**
Least scorer in MET i.e., Aditi must get call letter. This case is possible.
**Step 3:**
From condition (iii), scorer of 38 in CET, 56 in SAT, 51 in GAT, 60 in IFT must not get call letter.
From conditions (i) and (v); we can conclude that Charu and Dharmu must get call letter through IFT. And; now Charu can't get call letter through GAT in order to satisfy condition (iv). We can also see that;
Basil, Dharmu and Ekta must get call letter GAT. Now; Dharmu can't get call letter through CET and Ekta must receive call letter through CET but not through MET.
If Basil gets call letter through MET, then Aditi can't get call letter through both SAT and IFT. But in this case condition (ii) is violated.
Therefore, Basil had not received call letter through MET. And; Aditi had received admission call letter through either SAT or IFT.
Hence, the information can be shown in the following table.

| Test / Name | CET | SAT | GAT | IFT | MET | Total |
|---|---|---|---|---|---|---|
| Aditi | 38 x | 64 ✓ / x | 51 x | 76 x / ✓ | 82 ✓ | 2 |
| Basil | 42 ✓ | 56 x | 57 ✓ | 60 x | 86 x | 2 |
| Charu | 40 ✓ | 62 ✓ | 63 x | 62 ✓ | 88 ✓ | 4 |
| Dharmu | 45 x | 60 ✓ | 60 ✓ | 68 ✓ | 85 ✓ | 4 |
| Ekta | 41 ✓ | 59 ✓ | 65 ✓ | 72 ✓ | 94 x | 4 |
| Total | 3 | 4/3 | 3 | 3/4 | 3 | 16 |

**As per the table, required marks is 41.**

Bookmark FeedBack

**Directions for questions 5 to 10:** Answer the questions on the basis of the information given below.

Five aspirants - Aditi, Basil, Charu, Dharmu and Ekta appeared in five entrance tests - CET, SAT, GAT, IFT and MET - for admission in management colleges. Their marks in the above tests and their work experience while appearing for that test are as shown in the table given below:

| Name | Marks in test | | | | | Work experience (in years) while appearing | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CET | SAT | GAT | IFT | MET | CET | SAT | GAT | IFT | MET |
| Aditi | 38 | 64 | 51 | 76 | 82 | 1 | 2 | 2 | 1 | 1 |
| Basil | 42 | 56 | 57 | 60 | 86 | 1 | 2 | 3 | 1 | 4 |
| Charu | 40 | 62 | 63 | 62 | 88 | 2 | 1 | 2 | 2 | 3 |
| Dharmu | 45 | 60 | 60 | 68 | 85 | 1 | 1 | 1 | 4 | 2 |
| Ekta | 41 | 59 | 65 | 72 | 94 | 3 | 1 | 2 | 1 | 0 |

If an aspirant's marks in any test is equal to the average of the marks of two or three or four of the other aspirants and in addition his/her work experience while appearing for that test is the average of that of two or three or four of the other aspirants, then that aspirant will definitely receive a call letter for admission through that test. For the rest of the aspirants, the call letter may also based on other criteria apart from just their marks and the work experience.

**For example:-** If Aditi's marks in a test was an average of that of (2 others) Basil and Charu or (3 others) Basil, Charu and Dharmu or (4 others) Basil, Charu, Dharmu and Ekta and similarly work experience is also such an average, then Aditi must get a call letter to get admission through that test. If Aditi's marks does not satisfy this condition the other criteria will be considered for the call letter.

Some other known facts are as below:
(i) For every test there were either 3 or 4 aspirants who received a call letter.
(ii) Each aspirant received call letters through at least two tests.
(iii) Among all the five tests, only one aspirant with the lowest marks in a test received a call letter for admission.
(iv) No aspirant received call letters for admission through all five tests.
(v) If Aditi receives a call letter through a test, then Charu and Dharmu also receive call letters for the same test.
(vi) A total of 16 call letters were issued based on the above information.

**Q.7 [11831809]**
Through which of the following tests did Basil NOT receive a call letter?

1 ○ CET, IFT and MET

2 ○ SAT, GAT and MET

3 ◯ CET, GAT and MET

4 ◯ SAT, IFT and MET

**Solution:**
**Correct Answer : 4**

Answer key/Solution

**Step 1:**
First of all, we would pick those who must have received admission call letter based on information given in the question (Average based condition for both marks and experience).
From this; we see that 8 call letters were issued as below:

| | CET | SAT | GAT | IFT | MET |
|---|---|---|---|---|---|
| Aditi | | | | | |
| Basil | 42 ✓ | | | | |
| Charu | 40 ✓ | 62 ✓ | | | 88 ✓ |
| Dharmu | | 60 ✓ | | | 85 ✓ |
| Ekta | | 59 ✓ | | 72 ✓ | |

**Step 2:**
From condition (vi), total 16 call letters were issued which is only possible in such a way that 4 letters through, 1 test and 3 through each remaining.
Now, from condition (iii),

**Case 1:**
Least scorer in CET i.e., Aditi must get call letter.
This case would not be possible because there would be 2 tests through which 4 call letters were issued which violates conditions (i) and (vi) together.
**Case 2:**
Least scorer in SAT i.e., Basil must get call letter.
This case is not possible as it violates condition (ii).
**Case 3:**
Least scorer in GAT i.e., Aditi must get call letter.
This case violates condition (v).
**Case 4:**
Least scorer in IFT i.e., Basil must get call letter.
This case also violates condition (v).
**Case 5:**
Least scorer in MET i.e., Aditi must get call letter. This case is possible.
**Step 3:**
From condition (iii), scorer of 38 in CET, 56 in SAT, 51 in GAT, 60 in IFT must not get call letter.
From conditions (i) and (v); we can conclude that Charu and Dharmu must get call letter through IFT. And; now Charu can't get call letter through GAT in order to satisfy condition (iv). We can also see that;
Basil, Dharmu and Ekta must get call letter GAT. Now; Dharmu can't get call letter through CET and Ekta must receive call letter through CET but not through MET.
If Basil gets call letter through MET, then Aditi can't get call letter through both SAT and IFT. But in this case condition (ii) is violated.
Therefore, Basil had not received call letter through MET. And; Aditi had received admission call letter through either SAT or IFT.
Hence, the information can be shown in the following table.

| Test / Name | CET | SAT | GAT | IFT | MET | Total |
|---|---|---|---|---|---|---|
| Aditi | 38 x | 64 ✓ / x | 51 x | 76 x / ✓ | 82 ✓ | 2 |
| Basil | 42 ✓ | 56 x | 57 ✓ | 60 x | 86 x | 2 |
| Charu | 40 ✓ | 62 ✓ | 63 x | 62 ✓ | 88 ✓ | 4 |
| Dharmu | 45 x | 60 ✓ | 60 ✓ | 68 ✓ | 85 ✓ | 4 |
| Ekta | 41 ✓ | 59 ✓ | 65 ✓ | 72 ✓ | 94 x | 4 |
| Total | 3 | 4/3 | 3 | 3/4 | 3 | 16 |

**Basil did not get call letter through SAT, IIFT and MET.**

Bookmark FeedBack

**Directions for questions 5 to 10:** Answer the questions on the basis of the information given below.

Five aspirants - Aditi, Basil, Charu, Dharmu and Ekta appeared in five entrance tests - CET, SAT, GAT, IFT and MET - for admission in management colleges. Their marks in the above tests and their work experience while appearing for that test are as shown in the table given below:

| Name | Marks in test | | | | | Work experience (in years) while appearing | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CET | SAT | GAT | IFT | MET | CET | SAT | GAT | IFT | MET |
| Aditi | 38 | 64 | 51 | 76 | 82 | 1 | 2 | 2 | 1 | 1 |
| Basil | 42 | 56 | 57 | 60 | 86 | 1 | 2 | 3 | 1 | 4 |
| Charu | 40 | 62 | 63 | 62 | 88 | 2 | 1 | 2 | 2 | 3 |
| Dharmu | 45 | 60 | 60 | 68 | 85 | 1 | 1 | 1 | 4 | 2 |
| Ekta | 41 | 59 | 65 | 72 | 94 | 3 | 1 | 2 | 1 | 0 |

If an aspirant's marks in any test is equal to the average of the marks of two or three or four of the other aspirants and in addition his/her work experience while appearing for that test is the average of that of two or three or four of the other aspirants, then that aspirant will definitely receive a call letter for admission through that test. For the rest of the aspirants, the call letter may also based on other criteria apart from just their marks and the work experience.

**For example:-** If Aditi's marks in a test was an average of that of (2 others) Basil and Charu or (3 others) Basil, Charu and Dharmu or (4 others) Basil, Charu, Dharmu and Ekta and similarly work experience is also such an average, then Aditi must get a call letter to get admission through that test. If Aditi's marks does not satisfy this condition the other criteria will be considered for the call letter.

Some other known facts are as below:
(i) For every test there were either 3 or 4 aspirants who received a call letter.
(ii) Each aspirant received call letters through at least two tests.
(iii) Among all the five tests, only one aspirant with the lowest marks in a test received a call letter for admission.
(iv) No aspirant received call letters for admission through all five tests.
(v) If Aditi receives a call letter through a test, then Charu and Dharmu also receive call letters for the same test.
(vi) A total of 16 call letters were issued based on the above information.

**Q.8 [11831809]**
What are the marks of an aspirant who received a call letter inspite of being the lowest scorer in a test?

**Solution:**
**Correct Answer : 82**

Answer key/Solution

**Step 1:**
First of all, we would pick those who must have received admission call letter based on information given in the question (Average based condition for both marks and experience).
From this; we see that 8 call letters were issued as below:

| | CET | SAT | GAT | IFT | MET |
|---|---|---|---|---|---|
| Aditi | | | | | |
| Basil | 42 ✓ | | | | |
| Charu | 40 ✓ | 62 ✓ | | | 88 ✓ |
| Dharmu | | 60 ✓ | | | 85 ✓ |
| Ekta | | 59 ✓ | | 72 ✓ | |

**Step 2:**
From condition (vi), total 16 call letters were issued which is only possible in such a way that 4 letters through, 1 test and 3 through each remaining.
Now, from condition (iii),

**Case 1:**
Least scorer in CET i.e., Aditi must get call letter.
This case would not be possible because there would be 2 tests through which 4 call letters were issued which violates conditions (i) and (vi) together.
**Case 2:**
Least scorer in SAT i.e., Basil must get call letter.
This case is not possible as it violates condition (ii).
**Case 3:**
Least scorer in GAT i.e., Aditi must get call letter.
This case violates condition (v).
**Case 4:**
Least scorer in IFT i.e., Basil must get call letter.
This case also violates condition (v).
**Case 5:**
Least scorer in MET i.e., Aditi must get call letter. This case is possible.
**Step 3:**
From condition (iii), scorer of 38 in CET, 56 in SAT, 51 in GAT, 60 in IFT must not get call letter.
From conditions (i) and (v); we can conclude that Charu and Dharmu must get call letter through IFT. And; now Charu can't get call letter through GAT in order to satisfy condition (iv). We can also see that;
Basil, Dharmu and Ekta must get call letter GAT. Now; Dharmu can't get call letter through CET and Ekta must receive call letter through CET but not through MET.
If Basil gets call letter through MET, then Aditi can't get call letter through both SAT and IFT. But in this case condition (ii) is violated.
Therefore, Basil had not received call letter through MET. And; Aditi had received admission call letter through either SAT or IFT.
Hence, the information can be shown in the following table.

| Test / Name | CET | SAT | GAT | IFT | MET | Total |
|---|---|---|---|---|---|---|
| Aditi | 38 x | 64 ✓ / x | 51 x | 76 x / ✓ | 82 ✓ | 2 |
| Basil | 42 ✓ | 56 x | 57 ✓ | 60 x | 86 x | 2 |
| Charu | 40 ✓ | 62 ✓ | 63 x | 62 ✓ | 88 ✓ | 4 |
| Dharmu | 45 x | 60 ✓ | 60 ✓ | 68 ✓ | 85 ✓ | 4 |
| Ekta | 41 ✓ | 59 ✓ | 65 ✓ | 72 ✓ | 94 x | 4 |
| Total | 3 | 4/3 | 3 | 3/4 | 3 | 16 |

**Aditi, the least scorer in MET, receaved a call letter.**
**Hence, required least marks is 82.**

Bookmark FeedBack

**Directions for questions 5 to 10:** Answer the questions on the basis of the information given below.

Five aspirants - Aditi, Basil, Charu, Dharmu and Ekta appeared in five entrance tests - CET, SAT, GAT, IFT and MET - for admission in management colleges. Their marks in the above tests and their work experience while appearing for that test are as shown in the table given below:

| Name | Marks in test | | | | | Work experience (in years) while appearing | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CET | SAT | GAT | IFT | MET | CET | SAT | GAT | IFT | MET |
| Aditi | 38 | 64 | 51 | 76 | 82 | 1 | 2 | 2 | 1 | 1 |
| Basil | 42 | 56 | 57 | 60 | 86 | 1 | 2 | 3 | 1 | 4 |
| Charu | 40 | 62 | 63 | 62 | 88 | 2 | 1 | 2 | 2 | 3 |
| Dharmu | 45 | 60 | 60 | 68 | 85 | 1 | 1 | 1 | 4 | 2 |
| Ekta | 41 | 59 | 65 | 72 | 94 | 3 | 1 | 2 | 1 | 0 |

If an aspirant's marks in any test is equal to the average of the marks of two or three or four of the other aspirants and in addition his/her work experience while appearing for that test is the average of that of two or three or four of the other aspirants, then that aspirant will definitely receive a call letter for admission through that test. For the rest of the aspirants, the call letter may also based on other criteria apart from just their marks and the work experience.

**For example:-** If Aditi's marks in a test was an average of that of (2 others) Basil and Charu or (3 others) Basil, Charu and Dharmu or (4 others) Basil, Charu, Dharmu and Ekta and similarly work experience is also such an average, then Aditi must get a call letter to get admission through that test. If Aditi's marks does not satisfy this condition the other criteria will be considered for the call letter.

Some other known facts are as below:
(i) For every test there were either 3 or 4 aspirants who received a call letter.
(ii) Each aspirant received call letters through at least two tests.
(iii) Among all the five tests, only one aspirant with the lowest marks in a test received a call letter for admission.
(iv) No aspirant received call letters for admission through all five tests.
(v) If Aditi receives a call letter through a test, then Charu and Dharmu also receive call letters for the same test.
(vi) A total of 16 call letters were issued based on the above information.

**Q.9 [11831809]**
How many aspirant(s) received a call letter through four out of the five entrance tests?

1 ◯ 1

2 ◯ 2

3 ◯ 3

4 ◯ Either (2) or (3).

**Solution:**
**Correct Answer : 3**

Answer key/Solution

**Step 1:**
First of all, we would pick those who must have received admission call letter based on information given in the question (Average based condition for both marks and experience).
From this; we see that 8 call letters were issued as below:

| | CET | SAT | GAT | IFT | MET |
|---|---|---|---|---|---|
| Aditi | | | | | |
| Basil | 42 ✓ | | | | |
| Charu | 40 ✓ | 62 ✓ | | | 88 ✓ |
| Dharmu | | 60 ✓ | | | 85 ✓ |
| Ekta | | 59 ✓ | | 72 ✓ | |

**Step 2:**
From condition (vi), total 16 call letters were issued which is only possible in such a way that 4 letters through, 1 test and 3 through each remaining.
Now, from condition (iii),

**Case 1:**
Least scorer in CET i.e., Aditi must get call letter.
This case would not be possible because there would be 2 tests through which 4 call letters were issued which violates conditions (i) and (vi) together.
**Case 2:**
Least scorer in SAT i.e., Basil must get call letter.
This case is not possible as it violates condition (ii).
**Case 3:**
Least scorer in GAT i.e., Aditi must get call letter.
This case violates condition (v).
**Case 4:**
Least scorer in IFT i.e., Basil must get call letter.
This case also violates condition (v).
**Case 5:**
Least scorer in MET i.e., Aditi must get call letter. This case is possible.
**Step 3:**
From condition (iii), scorer of 38 in CET, 56 in SAT, 51 in GAT, 60 in IFT must not get call letter.
From conditions (i) and (v); we can conclude that Charu and Dharmu must get call letter through IFT. And; now Charu can't get call letter through GAT in order to satisfy condition (iv). We can also see that;
Basil, Dharmu and Ekta must get call letter GAT. Now; Dharmu can't get call letter through CET and Ekta must receive call letter through CET but not through MET.
If Basil gets call letter through MET, then Aditi can't get call letter through both SAT and IFT. But in this case condition (ii) is violated.
Therefore, Basil had not received call letter through MET. And; Aditi had received admission call letter through either SAT or IFT.
Hence, the information can be shown in the following table.

| Test / Name | CET | SAT | GAT | IFT | MET | Total |
|---|---|---|---|---|---|---|
| Aditi | 38 x | 64 ✓ / x | 51 x | 76 x / ✓ | 82 ✓ | 2 |
| Basil | 42 ✓ | 56 x | 57 ✓ | 60 x | 86 x | 2 |
| Charu | 40 ✓ | 62 ✓ | 63 x | 62 ✓ | 88 ✓ | 4 |
| Dharmu | 45 x | 60 ✓ | 60 ✓ | 68 ✓ | 85 ✓ | 4 |
| Ekta | 41 ✓ | 59 ✓ | 65 ✓ | 72 ✓ | 94 x | 4 |
| Total | 3 | 4/3 | 3 | 3/4 | 3 | 16 |

**Charu, Dharmu and Ekta receaved the call letters through 4 tests.**

Bookmark FeedBack

**Directions for questions 5 to 10:** Answer the questions on the basis of the information given below.

Five aspirants - Aditi, Basil, Charu, Dharmu and Ekta appeared in five entrance tests - CET, SAT, GAT, IFT and MET - for admission in management colleges. Their marks in the above tests and their work experience while appearing for that test are as shown in the table given below:

| Name | Marks in test | | | | | Work experience (in years) while appearing | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | CET | SAT | GAT | IFT | MET | CET | SAT | GAT | IFT | MET |
| Aditi | 38 | 64 | 51 | 76 | 82 | 1 | 2 | 2 | 1 | 1 |
| Basil | 42 | 56 | 57 | 60 | 86 | 1 | 2 | 3 | 1 | 4 |
| Charu | 40 | 62 | 63 | 62 | 88 | 2 | 1 | 2 | 2 | 3 |
| Dharmu | 45 | 60 | 60 | 68 | 85 | 1 | 1 | 1 | 4 | 2 |
| Ekta | 41 | 59 | 65 | 72 | 94 | 3 | 1 | 2 | 1 | 0 |

If an aspirant's marks in any test is equal to the average of the marks of two or three or four of the other aspirants and in addition his/her work experience while appearing for that test is the average of that of two or three or four of the other aspirants, then that aspirant will definitely receive a call letter for admission through that test. For the rest of the aspirants, the call letter may also based on other criteria apart from just their marks and the work experience.

**For example:-** If Aditi's marks in a test was an average of that of (2 others) Basil and Charu or (3 others) Basil, Charu and Dharmu or (4 others) Basil, Charu, Dharmu and Ekta and similarly work experience is also such an average, then Aditi must get a call letter to get admission through that test. If Aditi's marks does not satisfy this condition the other criteria will be considered for the call letter.

Some other known facts are as below:
(i) For every test there were either 3 or 4 aspirants who received a call letter.
(ii) Each aspirant received call letters through at least two tests.
(iii) Among all the five tests, only one aspirant with the lowest marks in a test received a call letter for admission.
(iv) No aspirant received call letters for admission through all five tests.
(v) If Aditi receives a call letter through a test, then Charu and Dharmu also receive call letters for the same test.
(vi) A total of 16 call letters were issued based on the above information.

**Q.10 [11831809]**
Through how many test(s) did both Aditi and Basil receive call letters?

1 ◯ 0

2 ◯ 1

3 ◯ 2

4 ◯ Either (2) or (3)

**Solution:**
**Correct Answer : 1**

Answer key/Solution

**Step 1:**
First of all, we would pick those who must have received admission call letter based on information given in the question (Average based condition for both marks and experience).
From this; we see that 8 call letters were issued as below:

| | CET | SAT | GAT | IFT | MET |
|---|---|---|---|---|---|
| Aditi | | | | | |
| Basil | 42 ✓ | | | | |
| Charu | 40 ✓ | 62 ✓ | | | 88 ✓ |
| Dharmu | | 60 ✓ | | | 85 ✓ |
| Ekta | | 59 ✓ | | 72 ✓ | |

**Step 2:**
From condition (vi), total 16 call letters were issued which is only possible in such a way that 4 letters through, 1 test and 3 through each remaining.
Now, from condition (iii),

**Case 1:**
Least scorer in CET i.e., Aditi must get call letter.
This case would not be possible because there would be 2 tests through which 4 call letters were issued which violates conditions (i) and (vi) together.
**Case 2:**
Least scorer in SAT i.e., Basil must get call letter.
This case is not possible as it violates condition (ii).
**Case 3:**
Least scorer in GAT i.e., Aditi must get call letter.
This case violates condition (v).
**Case 4:**
Least scorer in IFT i.e., Basil must get call letter.
This case also violates condition (v).
**Case 5:**
Least scorer in MET i.e., Aditi must get call letter. This case is possible.
**Step 3:**
From condition (iii), scorer of 38 in CET, 56 in SAT, 51 in GAT, 60 in IFT must not get call letter.
From conditions (i) and (v); we can conclude that Charu and Dharmu must get call letter through IFT. And; now Charu can't get call letter through GAT in order to satisfy condition (iv). We can also see that;
Basil, Dharmu and Ekta must get call letter GAT. Now; Dharmu can't get call letter through CET and Ekta must receive call letter through CET but not through MET.
If Basil gets call letter through MET, then Aditi can't get call letter through both SAT and IFT. But in this case condition (ii) is violated.
Therefore, Basil had not received call letter through MET. And; Aditi had received admission call letter through either SAT or IFT.
Hence, the information can be shown in the following table.

| Test / Name | CET | SAT | GAT | IFT | MET | Total |
|---|---|---|---|---|---|---|
| Aditi | 38 x | 64 ✓ / x | 51 x | 76 x / ✓ | 82 ✓ | 2 |
| Basil | 42 ✓ | 56 x | 57 ✓ | 60 x | 86 x | 2 |
| Charu | 40 ✓ | 62 ✓ | 63 x | 62 ✓ | 88 ✓ | 4 |
| Dharmu | 45 x | 60 ✓ | 60 ✓ | 68 ✓ | 85 ✓ | 4 |
| Ekta | 41 ✓ | 59 ✓ | 65 ✓ | 72 ✓ | 94 x | 4 |
| Total | 3 | 4/3 | 3 | 3/4 | 3 | 16 |

**We can't say with surety that both Aditi and Basil had received the call letters through any test.**

Bookmark FeedBack

**Directions for questions 11 to 16:** Answer the questions on the basis of the information given below.

A delivery boy has a delivery auto with shelves. He arranges his delivery boxes according to their weights in four shelves arranged vertically top to bottom - A, B, C and D. Each of the shelves has four compartments - I, II, III and IV- arranged in horizontally. Upon removing a box from a lower shelve for delivery, the box on the shelve above is moved down, such that no occupied compartment has any vacant compartments immediately below it. There are delivery boxes numbered from 1 to 16 having distinct weights in an Arithmetic Progression from 0.5 kg to 8 kg in any order. A permissible weight of any box in a particular shelve is mentioned in the bracket adjacent to the name of shelve.

The table has incomplete information regarding the boxes and their weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | - | - | 1 kg | 14 | 2 kg | - | - |
| B (< 4.5 kg) | - | 4 kg | 13 | - | 16 | - | 2 | 3 kg |
| C (< 6.5 kg) | - | 6 kg | - | - | 8 | 5 kg | - | 4.5 kg |
| D (< 8.5 kg) | 1 | - | 6 | 7 kg | - | 6.5 kg | 11 | – |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

(i) Box 15 is not on any row below Box 2 whereas Box 4 is placed in some row below Box 13.
(ii) Box 3, Box 5 and Box 9 are placed in the same shelve and Box 5 is 1 kg heavier than Box 9.
(iii) Box 10 has more weight than boxes 13 and 16.
(iv) The absolute difference between the box number of exactly one of the boxes, say Box K and its weight is 3 such that its weight is 7 kg more than that of Box (K + 1).
(v) The first delivery consists of Boxes 6, 9 and 12.
(vi) The second delivery consists of Boxes 1, 3, 5 and 15.
(vii) The fourth (i.e., last) delivery consists of 5 boxes and its total weight is 2 kg more than the total weight of the third delivery, which does not contain the box with the maximum weight.

**Q.11 [11831809]**
Which of the following combinations of box and weight is **NOT** correct?

1 ◯ Box 7 - 1.5 kg

2 ◯ Box 11 - 8 kg

3 ◯ Box 4 - 5.5 kg

4 ◯ Box 13 - 3.5 kg

**Solution:**
**Correct Answer : 3**

Answer key/Solution

**Step 1:**
Since all the weights are in AP from 0.5 to 8 kg. We have 16 distinct weights with a common difference of 0.5 kg. The top shelve A contains boxes with weights 0.5, 1, 1.5 and 2 kg.
Similarly, shelve B contains 2.5, 3, 3.5 and 4 kg ; shelve C contains 4.5, 5, 5.5 and 6 kg and shelve D contains 6.5, 7, 7.5 and 8 kg.
From condition (ii), we can say that Box 3, 5 and 9 are placed in shelve C as there are three places vacant for box numbers only in C. Also, Box 5 has 5.5 kg and Box 9 has 4.5 kg.
From condition (iv), we can conclude K = 11 and its weight is 8 kg whereas Box (K + 1) is Box 12 and its weight is 1 kg.
From condition (iii), we can say that Box 10 is kept in compartment I of shelve B.
From condition (i), we can say that Box 15 is placed in compartment IV of shelve A. The only box number remaining is Box 4, which is kept in compartment III of shelve D.
From the total weights given in the last row of the table and the information regarding the maximum weights that can be kept in any shelve, we can calculate the missing values of weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | 12 | 1 kg | 14 | 2 kg | 15 | 0.5 kg |
| B (< 4.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 16 | 2.5 kg | 2 | 3 kg |
| C (< 6.5 kg) | 3 | 6 kg | 5 | 5.5 kg | 8 | 5 kg | 9 | 4.5 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 6 | 7 kg | 4 | 6.5 kg | 11 | 8 kg |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

**Step 2:**
After the first delivery the shelves look like this :

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | | | 14 | 2 kg | | |
| B (< 4.5 kg) | 10 | 4 kg | | | 16 | 2.5 kg | 15 | 0.5 kg |
| C (< 6.5 kg) | 3 | 6 kg | 13 | 3.5 kg | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 5 | 5.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 3:**
After the second delivery the shelves looks like this:

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | | | | | 14 | 2 kg | | |
| B (< 4.5 kg) | | | | | 16 | 2.5 kg | | |
| C (< 6.5 kg) | 7 | 1.5 kg | | | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 4:**
Remaining total weight is 36 kg. Fourth (i.e., last) delivery is 2 kg more than the third delivery. Hence, 4th delivery is 19 kg (5 boxes) and the 3rd delivery is 17 kg (4 boxes, excluding Box 11).

**From the first table it is clear that option (3) is correct.**

Bookmark FeedBack

**Directions for questions 11 to 16:** Answer the questions on the basis of the information given below.

A delivery boy has a delivery auto with shelves. He arranges his delivery boxes according to their weights in four shelves arranged vertically top to bottom - A, B, C and D. Each of the shelves has four compartments - I, II, III and IV- arranged in horizontally. Upon removing a box from a lower shelve for delivery, the box on the shelve above is moved down, such that no occupied compartment has any vacant compartments immediately below it. There are delivery boxes numbered from 1 to 16 having distinct weights in an Arithmetic Progression from 0.5 kg to 8 kg in any order. A permissible weight of any box in a particular shelve is mentioned in the bracket adjacent to the name of shelve.

The table has incomplete information regarding the boxes and their weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | - | - | 1 kg | 14 | 2 kg | - | - |
| B (< 4.5 kg) | - | 4 kg | 13 | - | 16 | - | 2 | 3 kg |
| C (< 6.5 kg) | - | 6 kg | - | - | 8 | 5 kg | - | 4.5 kg |
| D (< 8.5 kg) | 1 | - | 6 | 7 kg | - | 6.5 kg | 11 | – |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

(i) Box 15 is not on any row below Box 2 whereas Box 4 is placed in some row below Box 13.
(ii) Box 3, Box 5 and Box 9 are placed in the same shelve and Box 5 is 1 kg heavier than Box 9.
(iii) Box 10 has more weight than boxes 13 and 16.
(iv) The absolute difference between the box number of exactly one of the boxes, say Box K and its weight is 3 such that its weight is 7 kg more than that of Box (K + 1).
(v) The first delivery consists of Boxes 6, 9 and 12.
(vi) The second delivery consists of Boxes 1, 3, 5 and 15.
(vii) The fourth (i.e., last) delivery consists of 5 boxes and its total weight is 2 kg more than the total weight of the third delivery, which does not contain the box with the maximum weight.

**Q.12 [11831809]**
Which of the following options represents the boxes in shelve D after the first delivery?

1 ◯ 1 - 5 - 4 - 11

2 ◯ 1 - 6 - 4 - 11

3 ◯ 7 - 13 - 8 - 11

4 ◯ 10 - 13 - 4 - 11

**Solution:**
**Correct Answer : 1**

Answer key/Solution

**Step 1:**
Since all the weights are in AP from 0.5 to 8 kg. We have 16 distinct weights with a common difference of 0.5 kg. The top shelve A contains boxes with weights 0.5, 1, 1.5 and 2 kg.
Similarly, shelve B contains 2.5, 3, 3.5 and 4 kg ; shelve C contains 4.5, 5, 5.5 and 6 kg and shelve D contains 6.5, 7, 7.5 and 8 kg.
From condition (ii), we can say that Box 3, 5 and 9 are placed in shelve C as there are three places vacant for box numbers only in C. Also, Box 5 has 5.5 kg and Box 9 has 4.5 kg.
From condition (iv), we can conclude K = 11 and its weight is 8 kg whereas Box (K + 1) is Box 12 and its weight is 1 kg.
From condition (iii), we can say that Box 10 is kept in compartment I of shelve B.
From condition (i), we can say that Box 15 is placed in compartment IV of shelve A. The only box number remaining is Box 4, which is kept in compartment III of shelve D.
From the total weights given in the last row of the table and the information regarding the maximum weights that can be kept in any shelve, we can calculate the missing values of weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | 12 | 1 kg | 14 | 2 kg | 15 | 0.5 kg |
| B (< 4.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 16 | 2.5 kg | 2 | 3 kg |
| C (< 6.5 kg) | 3 | 6 kg | 5 | 5.5 kg | 8 | 5 kg | 9 | 4.5 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 6 | 7 kg | 4 | 6.5 kg | 11 | 8 kg |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

**Step 2:**
After the first delivery the shelves look like this :

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | | | 14 | 2 kg | | |
| B (< 4.5 kg) | 10 | 4 kg | | | 16 | 2.5 kg | 15 | 0.5 kg |
| C (< 6.5 kg) | 3 | 6 kg | 13 | 3.5 kg | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 5 | 5.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 3:**
After the second delivery the shelves looks like this:

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | | | | | 14 | 2 kg | | |
| B (< 4.5 kg) | | | | | 16 | 2.5 kg | | |
| C (< 6.5 kg) | 7 | 1.5 kg | | | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 4:**
Remaining total weight is 36 kg. Fourth (i.e., last) delivery is 2 kg more than the third delivery. Hence, 4th delivery is 19 kg (5 boxes) and the 3rd delivery is 17 kg (4 boxes, excluding Box 11).

**It is clear from the second table that after the first delivery the boxes in shelve D are:**
**1 - 5 - 4 - 11.**

Bookmark FeedBack

**Directions for questions 11 to 16:** Answer the questions on the basis of the information given below.

A delivery boy has a delivery auto with shelves. He arranges his delivery boxes according to their weights in four shelves arranged vertically top to bottom - A, B, C and D. Each of the shelves has four compartments - I, II, III and IV- arranged in horizontally. Upon removing a box from a lower shelve for delivery, the box on the shelve above is moved down, such that no occupied compartment has any vacant compartments immediately below it. There are delivery boxes numbered from 1 to 16 having distinct weights in an Arithmetic Progression from 0.5 kg to 8 kg in any order. A permissible weight of any box in a particular shelve is mentioned in the bracket adjacent to the name of shelve.

The table has incomplete information regarding the boxes and their weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | - | - | 1 kg | 14 | 2 kg | - | - |
| B (< 4.5 kg) | - | 4 kg | 13 | - | 16 | - | 2 | 3 kg |
| C (< 6.5 kg) | - | 6 kg | - | - | 8 | 5 kg | - | 4.5 kg |
| D (< 8.5 kg) | 1 | - | 6 | 7 kg | - | 6.5 kg | 11 | – |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

(i) Box 15 is not on any row below Box 2 whereas Box 4 is placed in some row below Box 13.
(ii) Box 3, Box 5 and Box 9 are placed in the same shelve and Box 5 is 1 kg heavier than Box 9.
(iii) Box 10 has more weight than boxes 13 and 16.
(iv) The absolute difference between the box number of exactly one of the boxes, say Box K and its weight is 3 such that its weight is 7 kg more than that of Box (K + 1).
(v) The first delivery consists of Boxes 6, 9 and 12.
(vi) The second delivery consists of Boxes 1, 3, 5 and 15.
(vii) The fourth (i.e., last) delivery consists of 5 boxes and its total weight is 2 kg more than the total weight of the third delivery, which does not contain the box with the maximum weight.

**Q.13 [11831809]**
What is the difference (in g) in the total weight of boxes in shelve B and the total weight of boxes in shelve A after the second delivery?

**Solution:**
**Correct Answer : 500**

Answer key/Solution

**Step 1:**
Since all the weights are in AP from 0.5 to 8 kg. We have 16 distinct weights with a common difference of 0.5 kg. The top shelve A contains boxes with weights 0.5, 1, 1.5 and 2 kg.
Similarly, shelve B contains 2.5, 3, 3.5 and 4 kg ; shelve C contains 4.5, 5, 5.5 and 6 kg and shelve D contains 6.5, 7, 7.5 and 8 kg.
From condition (ii), we can say that Box 3, 5 and 9 are placed in shelve C as there are three places vacant for box numbers only in C. Also, Box 5 has 5.5 kg and Box 9 has 4.5 kg.
From condition (iv), we can conclude K = 11 and its weight is 8 kg whereas Box (K + 1) is Box 12 and its weight is 1 kg.
From condition (iii), we can say that Box 10 is kept in compartment I of shelve B.
From condition (i), we can say that Box 15 is placed in compartment IV of shelve A. The only box number remaining is Box 4, which is kept in compartment III of shelve D.
From the total weights given in the last row of the table and the information regarding the maximum weights that can be kept in any shelve, we can calculate the missing values of weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | 12 | 1 kg | 14 | 2 kg | 15 | 0.5 kg |
| B (< 4.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 16 | 2.5 kg | 2 | 3 kg |
| C (< 6.5 kg) | 3 | 6 kg | 5 | 5.5 kg | 8 | 5 kg | 9 | 4.5 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 6 | 7 kg | 4 | 6.5 kg | 11 | 8 kg |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

**Step 2:**
After the first delivery the shelves look like this :

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | | | 14 | 2 kg | | |
| B (< 4.5 kg) | 10 | 4 kg | | | 16 | 2.5 kg | 15 | 0.5 kg |
| C (< 6.5 kg) | 3 | 6 kg | 13 | 3.5 kg | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 5 | 5.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 3:**
After the second delivery the shelves looks like this:

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | | | | | 14 | 2 kg | | |
| B (< 4.5 kg) | | | | | 16 | 2.5 kg | | |
| C (< 6.5 kg) | 7 | 1.5 kg | | | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 4:**
Remaining total weight is 36 kg. Fourth (i.e., last) delivery is 2 kg more than the third delivery. Hence, 4th delivery is 19 kg (5 boxes) and the 3rd delivery is 17 kg (4 boxes, excluding Box 11).

**The box remaining in shelve A is Box 14 (2 kg) and that in shelve B is Box 16 (2.5 kg).**
**Hence, required difference in weight = 2.5 – 2 = 0.5 kg = 500 g.**

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**Directions for questions 11 to 16:** Answer the questions on the basis of the information given below.

A delivery boy has a delivery auto with shelves. He arranges his delivery boxes according to their weights in four shelves arranged vertically top to bottom - A, B, C and D. Each of the shelves has four compartments - I, II, III and IV- arranged in horizontally. Upon removing a box from a lower shelve for delivery, the box on the shelve above is moved down, such that no occupied compartment has any vacant compartments immediately below it. There are delivery boxes numbered from 1 to 16 having distinct weights in an Arithmetic Progression from 0.5 kg to 8 kg in any order. A permissible weight of any box in a particular shelve is mentioned in the bracket adjacent to the name of shelve.

The table has incomplete information regarding the boxes and their weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | - | - | 1 kg | 14 | 2 kg | - | - |
| B (< 4.5 kg) | - | 4 kg | 13 | - | 16 | - | 2 | 3 kg |
| C (< 6.5 kg) | - | 6 kg | - | - | 8 | 5 kg | - | 4.5 kg |
| D (< 8.5 kg) | 1 | - | 6 | 7 kg | - | 6.5 kg | 11 | – |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

(i) Box 15 is not on any row below Box 2 whereas Box 4 is placed in some row below Box 13.
(ii) Box 3, Box 5 and Box 9 are placed in the same shelve and Box 5 is 1 kg heavier than Box 9.
(iii) Box 10 has more weight than boxes 13 and 16.
(iv) The absolute difference between the box number of exactly one of the boxes, say Box K and its weight is 3 such that its weight is 7 kg more than that of Box (K + 1).
(v) The first delivery consists of Boxes 6, 9 and 12.
(vi) The second delivery consists of Boxes 1, 3, 5 and 15.
(vii) The fourth (i.e., last) delivery consists of 5 boxes and its total weight is 2 kg more than the total weight of the third delivery, which does not contain the box with the maximum weight.

**Q.14 [11831809]**
What is the ratio of the weights of all the boxes in the first and second deliveries respectively?

1 ◯ 23 : 37

2 ◯ 15 : 38

3 ◯ 25 : 39

4 ◯ 5 : 8

**Solution:**
**Correct Answer : 3**

Answer key/Solution

**Step 1:**
Since all the weights are in AP from 0.5 to 8 kg. We have 16 distinct weights with a common difference of 0.5 kg. The top shelve A contains boxes with weights 0.5, 1, 1.5 and 2 kg.
Similarly, shelve B contains 2.5, 3, 3.5 and 4 kg ; shelve C contains 4.5, 5, 5.5 and 6 kg and shelve D contains 6.5, 7, 7.5 and 8 kg.
From condition (ii), we can say that Box 3, 5 and 9 are placed in shelve C as there are three places vacant for box numbers only in C. Also, Box 5 has 5.5 kg and Box 9 has 4.5 kg.
From condition (iv), we can conclude K = 11 and its weight is 8 kg whereas Box (K + 1) is Box 12 and its weight is 1 kg.
From condition (iii), we can say that Box 10 is kept in compartment I of shelve B.
From condition (i), we can say that Box 15 is placed in compartment IV of shelve A. The only box number remaining is Box 4, which is kept in compartment III of shelve D.
From the total weights given in the last row of the table and the information regarding the maximum weights that can be kept in any shelve, we can calculate the missing values of weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | 12 | 1 kg | 14 | 2 kg | 15 | 0.5 kg |
| B (< 4.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 16 | 2.5 kg | 2 | 3 kg |
| C (< 6.5 kg) | 3 | 6 kg | 5 | 5.5 kg | 8 | 5 kg | 9 | 4.5 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 6 | 7 kg | 4 | 6.5 kg | 11 | 8 kg |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

**Step 2:**
After the first delivery the shelves look like this :

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | | | 14 | 2 kg | | |
| B (< 4.5 kg) | 10 | 4 kg | | | 16 | 2.5 kg | 15 | 0.5 kg |
| C (< 6.5 kg) | 3 | 6 kg | 13 | 3.5 kg | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 5 | 5.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 3:**
After the second delivery the shelves looks like this:

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | | | | | 14 | 2 kg | | |
| B (< 4.5 kg) | | | | | 16 | 2.5 kg | | |
| C (< 6.5 kg) | 7 | 1.5 kg | | | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 4:**
Remaining total weight is 36 kg. Fourth (i.e., last) delivery is 2 kg more than the third delivery. Hence, 4th delivery is 19 kg (5 boxes) and the 3rd delivery is 17 kg (4 boxes, excluding Box 11).

**We know that the first delivery consists of Boxes 6, 9 and 12 and the second delivery consists of Boxes 1, 3, 5 and 15.**
**Total weight of Boxes 6, 9 and 12 = 7 + 4.5 + 1 = 12.5 kg**

**Total weight of Boxes 1, 3, 5 and 15 = 7.5 + 6 + 5.5 + 0.5 = 19.5 kg**
**Hence, required ratio = 12.5 : 195 = 25 : 39.**

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**Directions for questions 11 to 16:** Answer the questions on the basis of the information given below.

A delivery boy has a delivery auto with shelves. He arranges his delivery boxes according to their weights in four shelves arranged vertically top to bottom - A, B, C and D. Each of the shelves has four compartments - I, II, III and IV- arranged in horizontally. Upon removing a box from a lower shelve for delivery, the box on the shelve above is moved down, such that no occupied compartment has any vacant compartments immediately below it. There are delivery boxes numbered from 1 to 16 having distinct weights in an Arithmetic Progression from 0.5 kg to 8 kg in any order. A permissible weight of any box in a particular shelve is mentioned in the bracket adjacent to the name of shelve.

The table has incomplete information regarding the boxes and their weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | - | - | 1 kg | 14 | 2 kg | - | - |
| B (< 4.5 kg) | - | 4 kg | 13 | - | 16 | - | 2 | 3 kg |
| C (< 6.5 kg) | - | 6 kg | - | - | 8 | 5 kg | - | 4.5 kg |
| D (< 8.5 kg) | 1 | - | 6 | 7 kg | - | 6.5 kg | 11 | – |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

(i) Box 15 is not on any row below Box 2 whereas Box 4 is placed in some row below Box 13.
(ii) Box 3, Box 5 and Box 9 are placed in the same shelve and Box 5 is 1 kg heavier than Box 9.
(iii) Box 10 has more weight than boxes 13 and 16.
(iv) The absolute difference between the box number of exactly one of the boxes, say Box K and its weight is 3 such that its weight is 7 kg more than that of Box (K + 1).
(v) The first delivery consists of Boxes 6, 9 and 12.
(vi) The second delivery consists of Boxes 1, 3, 5 and 15.
(vii) The fourth (i.e., last) delivery consists of 5 boxes and its total weight is 2 kg more than the total weight of the third delivery, which does not contain the box with the maximum weight.

**Q.15 [11831809]**
Which of these cannot be the combination of boxes kept in shelve D after the third delivery?

1 ◯ Box 13, Box 16, Box 11

2 ◯ Box 10, Box 13, Box 14, Box 11

3 ◯ Box 7, Box 13, Box 8, Box 11

4 ◯ Box 10, Box 16, Box 11

**Solution:**
**Correct Answer : 3**

Answer key/Solution

**Step 1:**
Since all the weights are in AP from 0.5 to 8 kg. We have 16 distinct weights with a common difference of 0.5 kg. The top shelve A contains boxes with weights 0.5, 1, 1.5 and 2 kg.
Similarly, shelve B contains 2.5, 3, 3.5 and 4 kg ; shelve C contains 4.5, 5, 5.5 and 6 kg and shelve D contains 6.5, 7, 7.5 and 8 kg.
From condition (ii), we can say that Box 3, 5 and 9 are placed in shelve C as there are three places vacant for box numbers only in C. Also, Box 5 has 5.5 kg and Box 9 has 4.5 kg.
From condition (iv), we can conclude K = 11 and its weight is 8 kg whereas Box (K + 1) is Box 12 and its weight is 1 kg.
From condition (iii), we can say that Box 10 is kept in compartment I of shelve B.
From condition (i), we can say that Box 15 is placed in compartment IV of shelve A. The only box number remaining is Box 4, which is kept in compartment III of shelve D.
From the total weights given in the last row of the table and the information regarding the maximum weights that can be kept in any shelve, we can calculate the missing values of weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | 12 | 1 kg | 14 | 2 kg | 15 | 0.5 kg |
| B (< 4.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 16 | 2.5 kg | 2 | 3 kg |
| C (< 6.5 kg) | 3 | 6 kg | 5 | 5.5 kg | 8 | 5 kg | 9 | 4.5 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 6 | 7 kg | 4 | 6.5 kg | 11 | 8 kg |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

**Step 2:**
After the first delivery the shelves look like this :

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | | | 14 | 2 kg | | |
| B (< 4.5 kg) | 10 | 4 kg | | | 16 | 2.5 kg | 15 | 0.5 kg |
| C (< 6.5 kg) | 3 | 6 kg | 13 | 3.5 kg | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 5 | 5.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 3:**
After the second delivery the shelves looks like this:

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | | | | | 14 | 2 kg | | |
| B (< 4.5 kg) | | | | | 16 | 2.5 kg | | |
| C (< 6.5 kg) | 7 | 1.5 kg | | | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 4:**
Remaining total weight is 36 kg. Fourth (i.e., last) delivery is 2 kg more than the third delivery. Hence, 4th delivery is 19 kg (5 boxes) and the 3rd delivery is 17 kg (4 boxes, excluding Box 11).

**In third delivery, 4 boxes with total weight of 17 kg will be delivered such that box 11 with weight 8 kg will not be delivered.**
**Different combinations of 17 kg can be:**

**Option (1): Box (7, 10, 4, 8): Here boxes left in shelve D will be 13, 16 and 11.**
**Option (2): Box (2, 4, 8, 16): Here boxes left in shelve D will be 10, 13, 14 and 11.**
**Option (4): Box (4, 8, 14, 13): Here boxes left in shelve D will be 10, 16 and 11.**
**So options (1), (2) and (4) are possible. Hence, option (3) is the correct answer.**

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**Directions for questions 11 to 16:** Answer the questions on the basis of the information given below.

A delivery boy has a delivery auto with shelves. He arranges his delivery boxes according to their weights in four shelves arranged vertically top to bottom - A, B, C and D. Each of the shelves has four compartments - I, II, III and IV- arranged in horizontally. Upon removing a box from a lower shelve for delivery, the box on the shelve above is moved down, such that no occupied compartment has any vacant compartments immediately below it. There are delivery boxes numbered from 1 to 16 having distinct weights in an Arithmetic Progression from 0.5 kg to 8 kg in any order. A permissible weight of any box in a particular shelve is mentioned in the bracket adjacent to the name of shelve.

The table has incomplete information regarding the boxes and their weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | - | - | 1 kg | 14 | 2 kg | - | - |
| B (< 4.5 kg) | - | 4 kg | 13 | - | 16 | - | 2 | 3 kg |
| C (< 6.5 kg) | - | 6 kg | - | - | 8 | 5 kg | - | 4.5 kg |
| D (< 8.5 kg) | 1 | - | 6 | 7 kg | - | 6.5 kg | 11 | – |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

(i) Box 15 is not on any row below Box 2 whereas Box 4 is placed in some row below Box 13.
(ii) Box 3, Box 5 and Box 9 are placed in the same shelve and Box 5 is 1 kg heavier than Box 9.
(iii) Box 10 has more weight than boxes 13 and 16.
(iv) The absolute difference between the box number of exactly one of the boxes, say Box K and its weight is 3 such that its weight is 7 kg more than that of Box (K + 1).
(v) The first delivery consists of Boxes 6, 9 and 12.
(vi) The second delivery consists of Boxes 1, 3, 5 and 15.
(vii) The fourth (i.e., last) delivery consists of 5 boxes and its total weight is 2 kg more than the total weight of the third delivery, which does not contain the box with the maximum weight.

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**Q.16 [11831809]**
Select the correct order of boxes in terms of their weights.

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1 ◯ Box 6 > Box 16 > Box 15 > Box 10

---

2 ◯ Box 11 > Box 7 > Box 3 > Box 1

---

3 ◯ Box 1 > Box 2 > Box 3 > Box 4

4 ◯ Box 4 > Box 5 > Box 13 > Box 15

**Solution:**
**Correct Answer : 4**

Answer key/Solution

**Step 1:**
Since all the weights are in AP from 0.5 to 8 kg. We have 16 distinct weights with a common difference of 0.5 kg. The top shelve A contains boxes with weights 0.5, 1, 1.5 and 2 kg.
Similarly, shelve B contains 2.5, 3, 3.5 and 4 kg ; shelve C contains 4.5, 5, 5.5 and 6 kg and shelve D contains 6.5, 7, 7.5 and 8 kg.
From condition (ii), we can say that Box 3, 5 and 9 are placed in shelve C as there are three places vacant for box numbers only in C. Also, Box 5 has 5.5 kg and Box 9 has 4.5 kg.
From condition (iv), we can conclude K = 11 and its weight is 8 kg whereas Box (K + 1) is Box 12 and its weight is 1 kg.
From condition (iii), we can say that Box 10 is kept in compartment I of shelve B.
From condition (i), we can say that Box 15 is placed in compartment IV of shelve A. The only box number remaining is Box 4, which is kept in compartment III of shelve D.
From the total weights given in the last row of the table and the information regarding the maximum weights that can be kept in any shelve, we can calculate the missing values of weights.

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | 12 | 1 kg | 14 | 2 kg | 15 | 0.5 kg |
| B (< 4.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 16 | 2.5 kg | 2 | 3 kg |
| C (< 6.5 kg) | 3 | 6 kg | 5 | 5.5 kg | 8 | 5 kg | 9 | 4.5 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 6 | 7 kg | 4 | 6.5 kg | 11 | 8 kg |
| Total | | 19 kg | | 17 kg | | 16 kg | | 16 kg |

**Step 2:**
After the first delivery the shelves look like this :

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | 7 | 1.5 kg | | | 14 | 2 kg | | |
| B (< 4.5 kg) | 10 | 4 kg | | | 16 | 2.5 kg | 15 | 0.5 kg |
| C (< 6.5 kg) | 3 | 6 kg | 13 | 3.5 kg | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 1 | 7.5 kg | 5 | 5.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 3:**
After the second delivery the shelves looks like this:

| Shelves (weight allowed) | I | | II | | III | | IV | |
|---|---|---|---|---|---|---|---|---|
| | Box No. | Weight | Box No. | Weight | Box No. | Weight | Box No. | Weight |
| A (< 2.5 kg) | | | | | 14 | 2 kg | | |
| B (< 4.5 kg) | | | | | 16 | 2.5 kg | | |
| C (< 6.5 kg) | 7 | 1.5 kg | | | 8 | 5 kg | 2 | 3 kg |
| D (< 8.5 kg) | 10 | 4 kg | 13 | 3.5 kg | 4 | 6.5 kg | 11 | 8 kg |

**Step 4:**
Remaining total weight is 36 kg. Fourth (i.e., last) delivery is 2 kg more than the third delivery. Hence, 4th delivery is 19 kg (5 boxes) and the 3rd delivery is 17 kg (4 boxes, excluding Box 11).

**We can see from the first table that option (4) is correct.**

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**Directions for questions 17 to 20:** Answer the questions on the basis of the information given below.

In 2021, 16 million cases were filed across 9 major cities in India of which 80% of the cases were reported under 6 major categories – A, B, C, D, E, and others. The table 1 given below shows the percentage distribution of these cases filed in the 9 major cities. The percentage for these cities was deleted by mistake. The table 2 given below shows the percentage distribution of crimes reported category wise within these 6 categories.
Further, it is also know that:
(i) Cases filed in Delhi were equivalent to 90% of cases filed in Chennai and 180% of cases filed in Mumbai.
(ii) 90% of total cases filed under categories D and E were reported only in four major cities Trivandrum, Kolkata, Delhi and Mumbai in the ratio 4 : 2 : 1 : 1.

**Table 1**

| City | Trivandrum | Kolkata | Hyderabad | Chennai | Delhi | Mumbai | Bangalore | Bhopal | Jaipur |
|---|---|---|---|---|---|---|---|---|---|
| Percentage distribution of the cases filed | 40% | 20% | 10% | 10% | – | – | – | 2% | 1.50% |

**Table 2**

| Category | A | B | C | D | E | Others |
|---|---|---|---|---|---|---|
| Percentage distribution categorywise | 8% | 10% | 5% | 15% | 30% | – |

**Q.17 [11831809]**
What is the total number of cases filed in Bangalore?

**Solution:**
**Correct Answer : 400000**

Answer key/Solution

**Step 1:**
From condition (i), Case filed in Delhi is equal to 14,40,000 and cases filed in Mumbai is equal to 8,00,000. Therefore, cases filed in Bangalore must be 4,00,000.
**Step 2:**
Complete information in Table-1 and Table-2 can be shown as:

Table-1

| City | Percentage distribution of the cases filed | Total cases |
|---|---|---|
| Trivandrum | 40% | 6400000 |
| Kolkata | 20% | 3200000 |
| Hyderabad | 10% | 1600000 |
| Chennai | 10% | 1600000 |
| Delhi | 9% | 1440000 |
| Mumbai | 5% | 800000 |
| Bangalore | 2.50% | 400000 |
| Bhopal | 2% | 320000 |
| Jaipur | 1.50% | 240000 |

Table-2

| Crime | Percentage distribution category wise | Cases |
|---|---|---|
| A | 8% | 1024000 |
| B | 10% | 1280000 |
| C | 5% | 640000 |
| D | 15% | 1920000 |
| E | 30% | 3840000 |
| Others | 32% | 4096000 |
| | | |
| | | |
| | | |

**4,00,000 cases are filed in Bangalore.**

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**Directions for questions 17 to 20:** Answer the questions on the basis of the information given below.

In 2021, 16 million cases were filed across 9 major cities in India of which 80% of the cases were reported under 6 major categories – A, B, C, D, E, and others. The table 1 given below shows the percentage distribution of these cases filed in the 9 major cities. The percentage for these cities was deleted by mistake. The table 2 given below shows the percentage distribution of crimes reported category wise within these 6 categories.
Further, it is also know that:
(i) Cases filed in Delhi were equivalent to 90% of cases filed in Chennai and 180% of cases filed in Mumbai.
(ii) 90% of total cases filed under categories D and E were reported only in four major cities Trivandrum, Kolkata, Delhi and Mumbai in the ratio 4 : 2 : 1 : 1.

**Table 1**

| City | Trivandrum | Kolkata | Hyderabad | Chennai | Delhi | Mumbai | Bangalore | Bhopal | Jaipur |
|---|---|---|---|---|---|---|---|---|---|
| Percentage distribution of the cases filed | 40% | 20% | 10% | 10% | – | – | – | 2% | 1.50% |

**Table 2**

| Category | A | B | C | D | E | Others |
|---|---|---|---|---|---|---|
| Percentage distribution categorywise | 8% | 10% | 5% | 15% | 30% | – |

**Q.18 [11831809]**
If crimes D and E were reported in the ratio of 3 : 2 in Kolkata, then what was the total number of cases filed in Kolkata under category D?

1 ◯ 777600

2 ◯ 518400

3 ◯ 648000

4 ◯ 1555200

**Solution:**
**Correct Answer : 1**

Answer key/Solution

**Step 1:**
From condition (i), Case filed in Delhi is equal to 14,40,000 and cases filed in Mumbai is equal to 8,00,000. Therefore, cases filed in Bangalore must be 4,00,000.
**Step 2:**
Complete information in Table-1 and Table-2 can be shown as:

Table-1

| City | Percentage distribution of the cases filed | Total cases |
|---|---|---|
| Trivandrum | 40% | 6400000 |
| Kolkata | 20% | 3200000 |
| Hyderabad | 10% | 1600000 |
| Chennai | 10% | 1600000 |
| Delhi | 9% | 1440000 |
| Mumbai | 5% | 800000 |
| Bangalore | 2.50% | 400000 |
| Bhopal | 2% | 320000 |
| Jaipur | 1.50% | 240000 |

Table-2

| Crime | Percentage distribution category wise | Cases |
|---|---|---|
| A | 8% | 1024000 |
| B | 10% | 1280000 |
| C | 5% | 640000 |
| D | 15% | 1920000 |
| E | 30% | 3840000 |
| Others | 32% | 4096000 |
| | | |
| | | |
| | | |

**Number of cases filed under categories D and E in Kolkata = 5760000 × 0.9 × (2/8) = 1296000**
**Hence, cases filed under category D = 1296000 × (3/5) = 777600.**

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**Directions for questions 17 to 20:** Answer the questions on the basis of the information given below.

In 2021, 16 million cases were filed across 9 major cities in India of which 80% of the cases were reported under 6 major categories – A, B, C, D, E, and others. The table 1 given below shows the percentage distribution of these cases filed in the 9 major cities. The percentage for these cities was deleted by mistake. The table 2 given below shows the percentage distribution of crimes reported category wise within these 6 categories.
Further, it is also know that:
(i) Cases filed in Delhi were equivalent to 90% of cases filed in Chennai and 180% of cases filed in Mumbai.
(ii) 90% of total cases filed under categories D and E were reported only in four major cities Trivandrum, Kolkata, Delhi and Mumbai in the ratio 4 : 2 : 1 : 1.

**Table 1**

| City | Trivandrum | Kolkata | Hyderabad | Chennai | Delhi | Mumbai | Bangalore | Bhopal | Jaipur |
|---|---|---|---|---|---|---|---|---|---|
| Percentage distribution of the cases filed | 40% | 20% | 10% | 10% | – | – | – | 2% | 1.50% |

**Table 2**

| Category | A | B | C | D | E | Others |
|---|---|---|---|---|---|---|
| Percentage distribution categorywise | 8% | 10% | 5% | 15% | 30% | – |

**Q.19 [11831809]**
If 4%, 9%, 10%, 5% of cases in Trivandrum, Kolkata, Hyderabad and Chennai respectively were filed under categories A, then which of the following is the correct order of cities regarding the number of cases filed?

1 ◯ Trivandrum > Kolkata > Hyderabad > Chennai

2 ◯ Kolkata > Hyderabad > Trivandrum > Chennai

3 ◯ Kolkata > Trivandrum > Chennai > Hyderabad

4 ◯ Kolkata > Trivandrum > Hyderabad > Chennai

**Solution:**
**Correct Answer : 4**

Answer key/Solution

**Step 1:**
From condition (i), Case filed in Delhi is equal to 14,40,000 and cases filed in Mumbai is equal to 8,00,000. Therefore, cases filed in Bangalore must be 4,00,000.
**Step 2:**
Complete information in Table-1 and Table-2 can be shown as:

Table-1

| City | Percentage distribution of the cases filed | Total cases |
|---|---|---|
| Trivandrum | 40% | 6400000 |
| Kolkata | 20% | 3200000 |
| Hyderabad | 10% | 1600000 |
| Chennai | 10% | 1600000 |
| Delhi | 9% | 1440000 |
| Mumbai | 5% | 800000 |
| Bangalore | 2.50% | 400000 |
| Bhopal | 2% | 320000 |
| Jaipur | 1.50% | 240000 |

Table-2

| Crime | Percentage distribution category wise | Cases |
|---|---|---|
| A | 8% | 1024000 |
| B | 10% | 1280000 |
| C | 5% | 640000 |
| D | 15% | 1920000 |
| E | 30% | 3840000 |
| Others | 32% | 4096000 |
| | | |
| | | |
| | | |

**4%, 9%, 10%, 5% of cases in Trivandrum, Kolkata, Hyderabad and Chennai respectively were filed under categories**
**A which is equivalent to 256000, 288000, 160000, and 80000 respectively.**
**Hence, the correct order is Kolkata > Trivandrum > Hyderabad > Chennai.**

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**Directions for questions 17 to 20:** Answer the questions on the basis of the information given below.

In 2021, 16 million cases were filed across 9 major cities in India of which 80% of the cases were reported under 6 major categories – A, B, C, D, E, and others. The table 1 given below shows the percentage distribution of these cases filed in the 9 major cities. The percentage for these cities was deleted by mistake. The table 2 given below shows the percentage distribution of crimes reported category wise within these 6 categories.
Further, it is also know that:
(i) Cases filed in Delhi were equivalent to 90% of cases filed in Chennai and 180% of cases filed in Mumbai.
(ii) 90% of total cases filed under categories D and E were reported only in four major cities Trivandrum, Kolkata, Delhi and Mumbai in the ratio 4 : 2 : 1 : 1.

**Table 1**

| City | Trivandrum | Kolkata | Hyderabad | Chennai | Delhi | Mumbai | Bangalore | Bhopal | Jaipur |
|---|---|---|---|---|---|---|---|---|---|
| Percentage distribution of the cases filed | 40% | 20% | 10% | 10% | – | – | – | 2% | 1.50% |

**Table 2**

| Category | A | B | C | D | E | Others |
|---|---|---|---|---|---|---|
| Percentage distribution categorywise | 8% | 10% | 5% | 15% | 30% | – |

**Q.20 [11831809]**

60% and 80% of crimes in Bhopal and Jaipur were from category “others”. The rest of the cases filed in Bhopal and Jaipur belonged to the categories A, B and C in the ratio 1 : 1 : 2 and 1 : 2 : 3, in that order. What is the absolute difference between the cases filed in category C in cities Bhopal and Jaipur?

**Solution:**
**Correct Answer : 40000**

Answer key/Solution

**Step 1:**
From condition (i), Case filed in Delhi is equal to 14,40,000 and cases filed in Mumbai is equal to 8,00,000. Therefore, cases filed in Bangalore must be 4,00,000.
**Step 2:**
Complete information in Table-1 and Table-2 can be shown as:

Table-1

| City | Percentage distribution of the cases filed | Total cases |
|---|---|---|
| Trivandrum | 40% | 6400000 |
| Kolkata | 20% | 3200000 |
| Hyderabad | 10% | 1600000 |
| Chennai | 10% | 1600000 |
| Delhi | 9% | 1440000 |
| Mumbai | 5% | 800000 |
| Bangalore | 2.50% | 400000 |
| Bhopal | 2% | 320000 |
| Jaipur | 1.50% | 240000 |

Table-2

| Crime | Percentage distribution category wise | Cases |
|---|---|---|
| A | 8% | 1024000 |
| B | 10% | 1280000 |
| C | 5% | 640000 |
| D | 15% | 1920000 |
| E | 30% | 3840000 |
| Others | 32% | 4096000 |
| | | |
| | | |
| | | |

**60% and 80% of crimes in Bhopal and Jaipur were from other category crimes which is equivalent to 192000 and 192000.**
**Rest of the cases in Bhopal = 128000**
**Rest of the cases in Jaipur = 48000**
**Hence, required difference = 128000 × (2/4) – 48000 × (3/6) = 40,000.**

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